HUMBOLDT-UNIVERSITÄT ZU BERLIN

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Models of Neural Systems, WS 2024/25
Project 2: Sharp Waves in a Bistable Network
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Introduction During deep sleep, sudden bursts of activity arise in the otherwise almost quiescent hippocampus. They are called sharp waves (SPWs, or SWR, where the R stands for ripples) and they have been linked to memory consolidation, because they coincide with the replay of waking activity (Diba and Buzsáki, 2007) and because their disruption impairs performance in cognitive tasks involving newly acquired memories (Girardeau et al., 2009). Both pyramidal cells (P) and PV-basket cells (B), a class of interneurons, largely increase their firing rate during SPWs (Csicsvari et al., 1999). Surprisingly, stimulation of B cells has been shown to induce such SPWs, a paradoxical excitation-by-inhibition effect (Schlingloff et al., 2014).

A possible explanation for this effect has been advanced by Evangelista et al. (2020) by suggesting that B cells disinhibit P cells by inhibiting a different class of "anti-sharp-wave" (A) interneurons. The latter would be hypothetical cells which tonically fire during non-SPW times and keep the other populations mostly suppressed. Activation of B cells, either induced or spontaneously occurring due to noise, would instead suppress A cells and switch the network to the SPW state. SPWs would terminate, for example, due to synaptic depression, a form of short-term plasticity.

Steps

1. Read Evangelista et al. (2020) and familiarize yourself with the context, methods, and results of this paper. Focus primarily on the spiking model, since the rate model is not included in this project.

2. Using Brian (www.briansimulator.org), implement the network of integrate-and-fire neurons described in the Methods (Equations 1-3, exclude Equation 4).

3. Block synaptic depression to first examine the steady-state landscape of the network without short-term plasticity. Deliver positive and negative stimulation pulses to population B (or to other population) and observe how the network switches from one state to the other. Does this look like Figure 2?

4. Now introduce synaptic depression and observe the spontaneously occurring sharp waves. Compare their dynamics to Figures 3 and 9 and think how to quantify their features, such as their incidence and duration, the peak frequency in the power spectrum, the duration of the inter-event intervals, and the average firing rate of each

population in each of the two states. For this, you can either follow the LFP estimation procedure described in the paper (Section "quantification of SWR properties") or define your own criteria directly based on firing rates.

5. In the paper, special attention is given to the correlation between the duration of the inter-event intervals and the size of the following sharp wave (Figure 11). Do you see the same effect? (It is enough to quantify this for spontaneous events).

6. **(Extra)** If all your results so far are consistent with the original paper and you still have time, you can try and vary one of the parameters which were left untouched in the model: the constant background current of 200 pA received by all the populations. You can vary this in the 100–300 pA range and observe how this affects the incidence and statistics of SPWs. Do you think this is a universal property, or does it depend on the specific choice of the other parameters?

Create labeled figures for all your main findings and summarize your results in a brief report.

References

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